

Bayesian Graphical Modeling

Network Comparisons

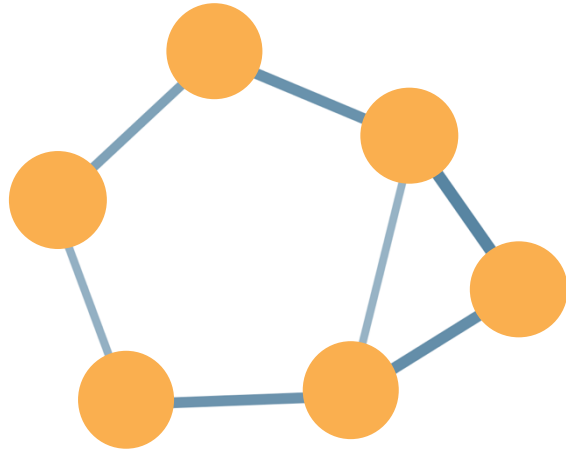
PsyNets Winterschool 2026

m.marsman@uva.nl

bayesiangraphicalmodeling.com

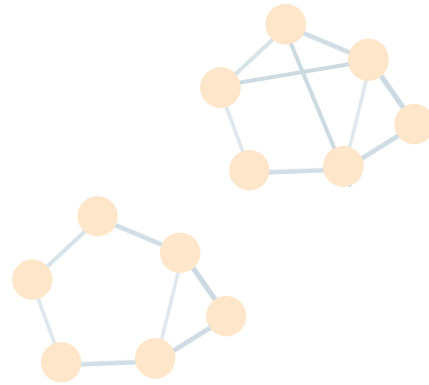


Setup Today

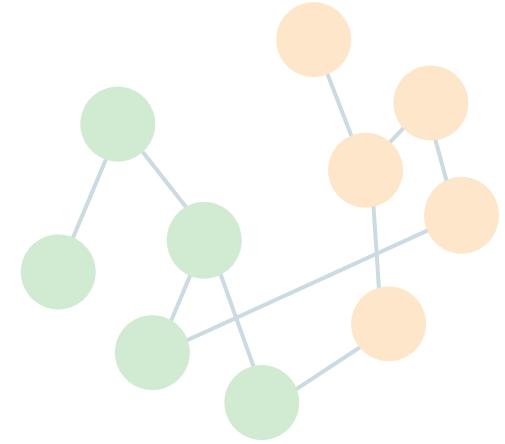


Estimating & Testing

Morning Session



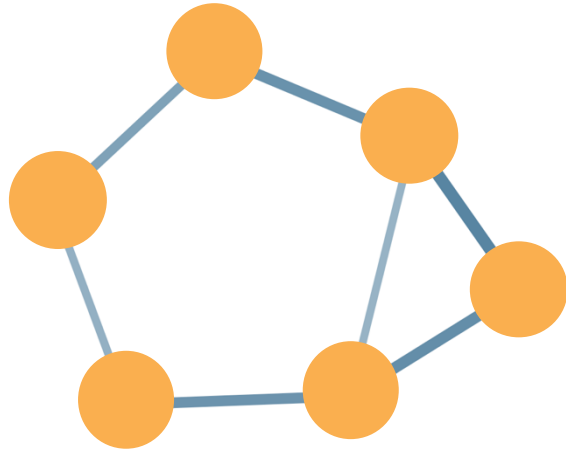
Group Comparison



Clustering

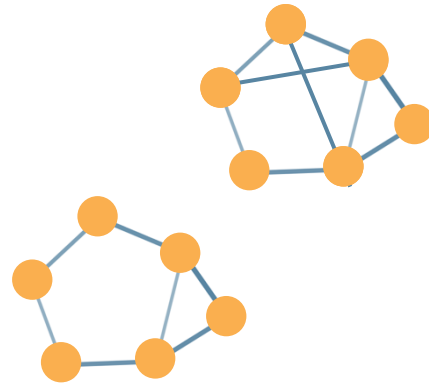
Afternoon Session

Setup Today



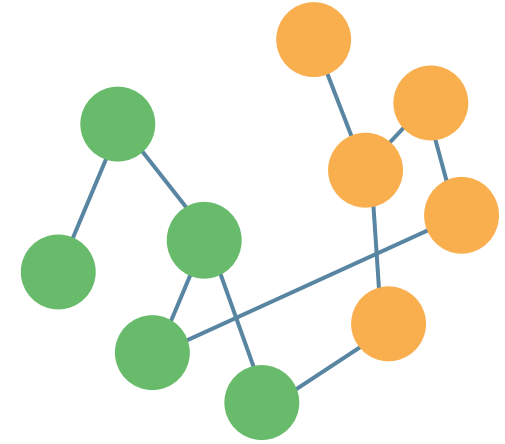
Estimating & Testing

Morning Session



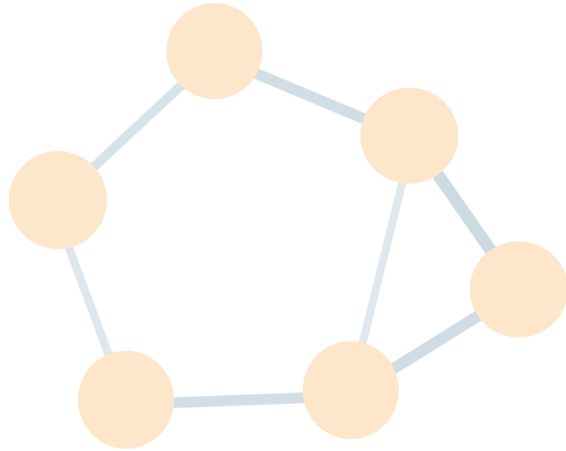
Group Comparison

Afternoon Session



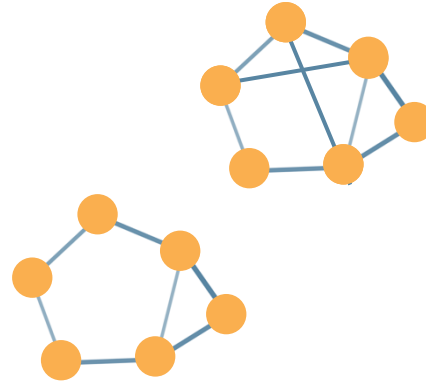
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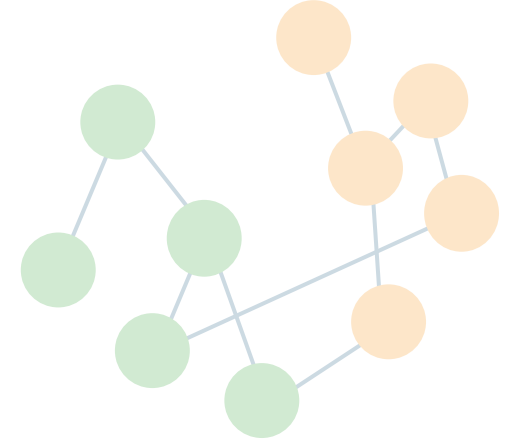
Estimating & Testing

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Group Comparison

Afternoon Session



Clustering

Session setup



Part I: Theory



Part II: Tutorial / Practical

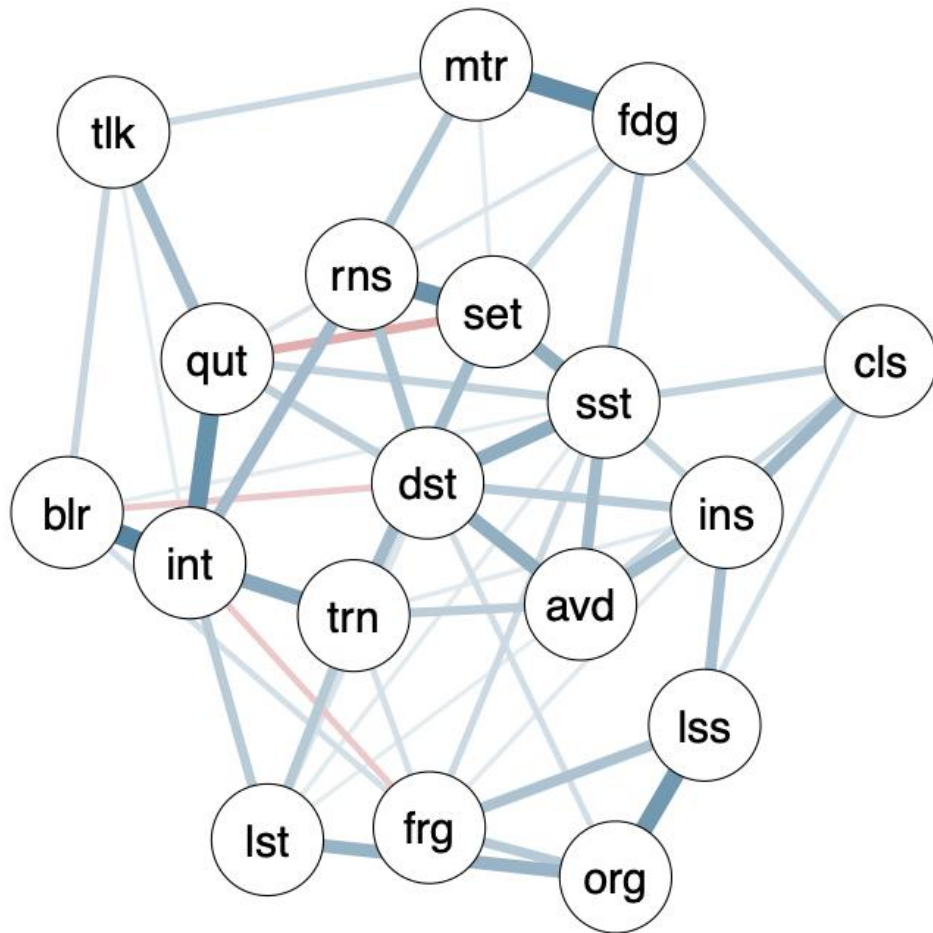


Session setup



Part I: Theory

A Network of ADHD Variables



Nodes are ADHD symptoms (present/absent).

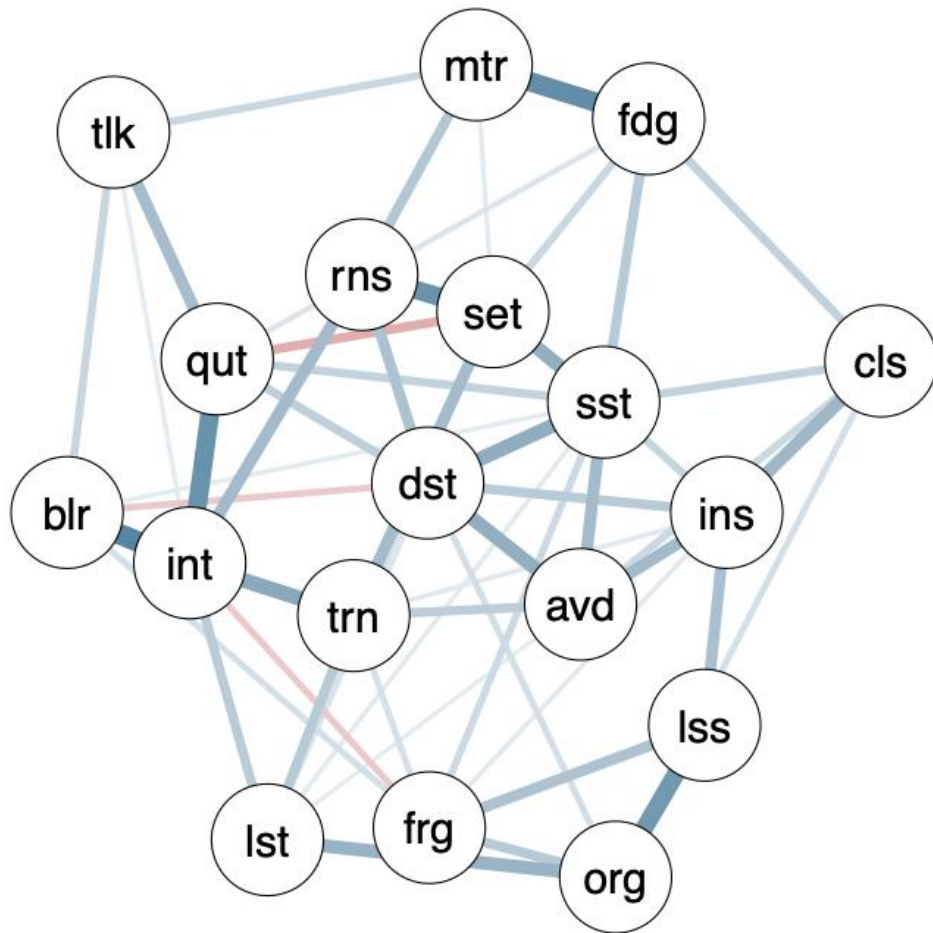
Edges are partial associations.

Goal is to determine conditional dependence relationships.

library(bgms)
?ADHD

Silk, et al. (2019). A network analysis approach to ADHD symptoms: More than the sum of its parts. *PLOS ONE*, 14(1), e0211053.

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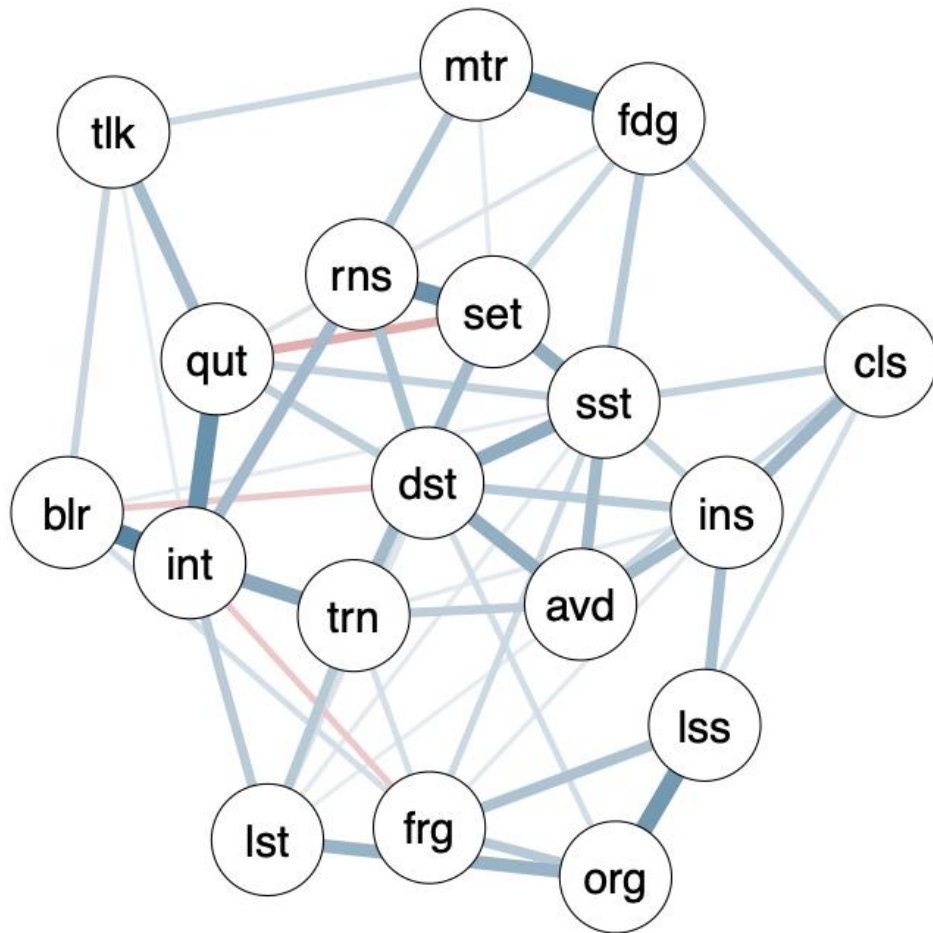
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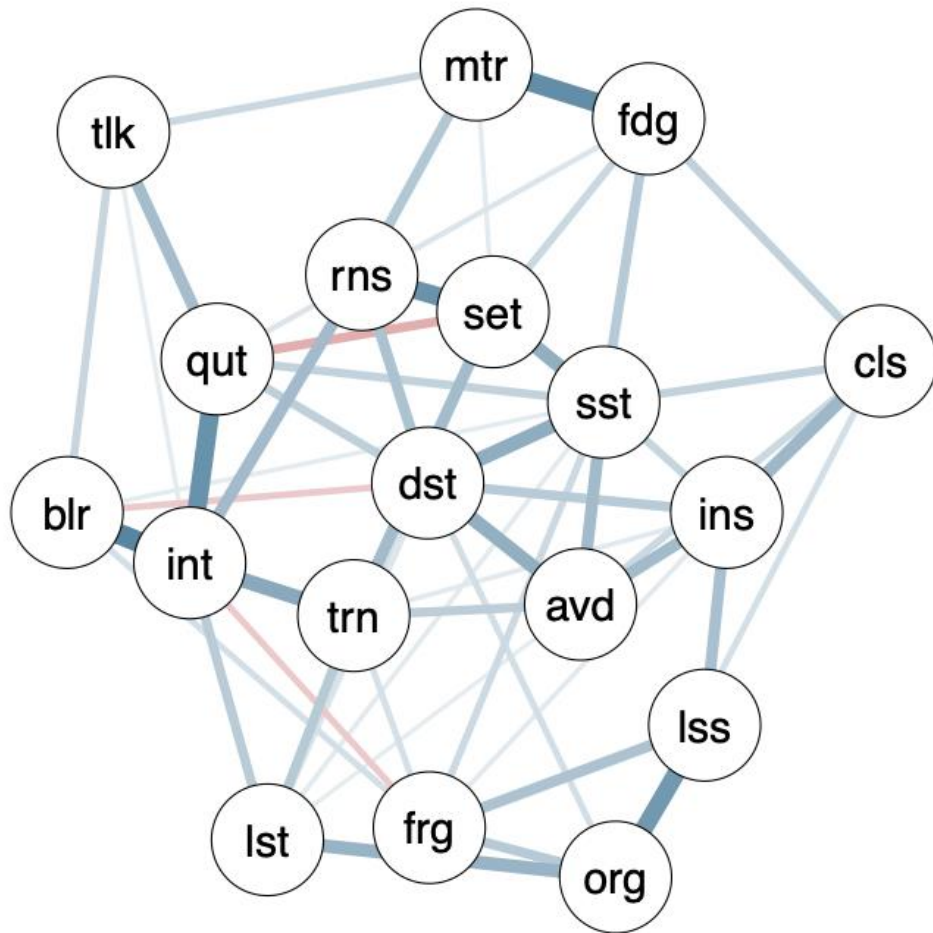
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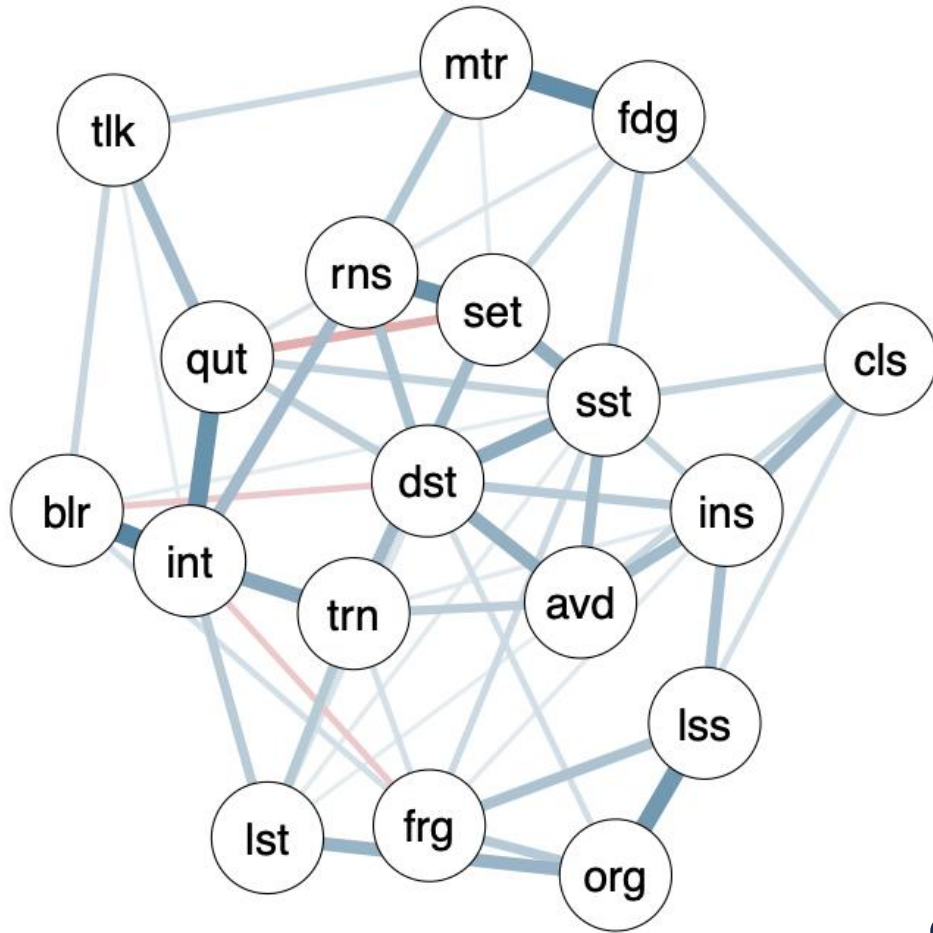
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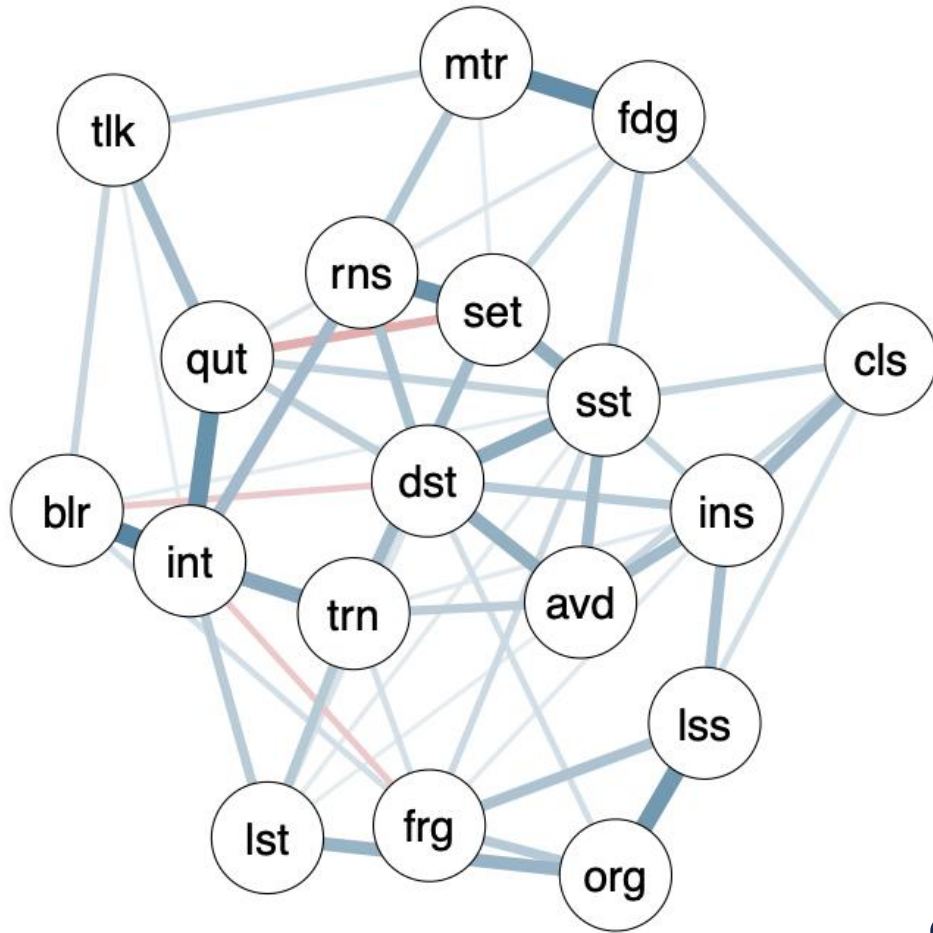
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Ising Model or Ordinal Markov Random Field



$$P(\mathbf{X} = \mathbf{x}) = \frac{\exp \left(\sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left(\sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

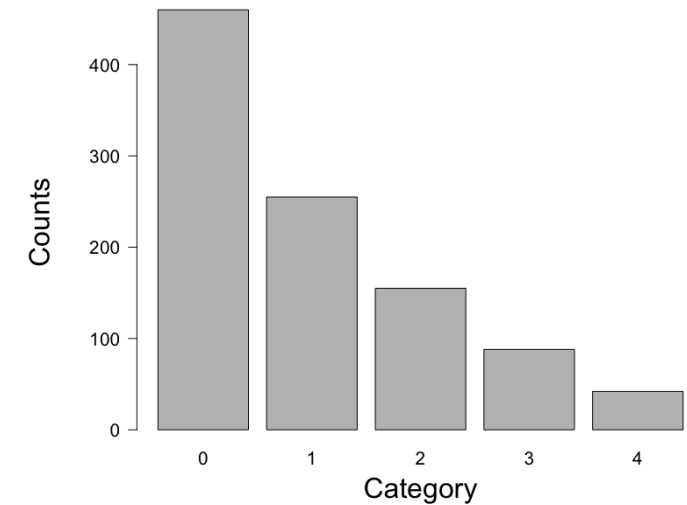
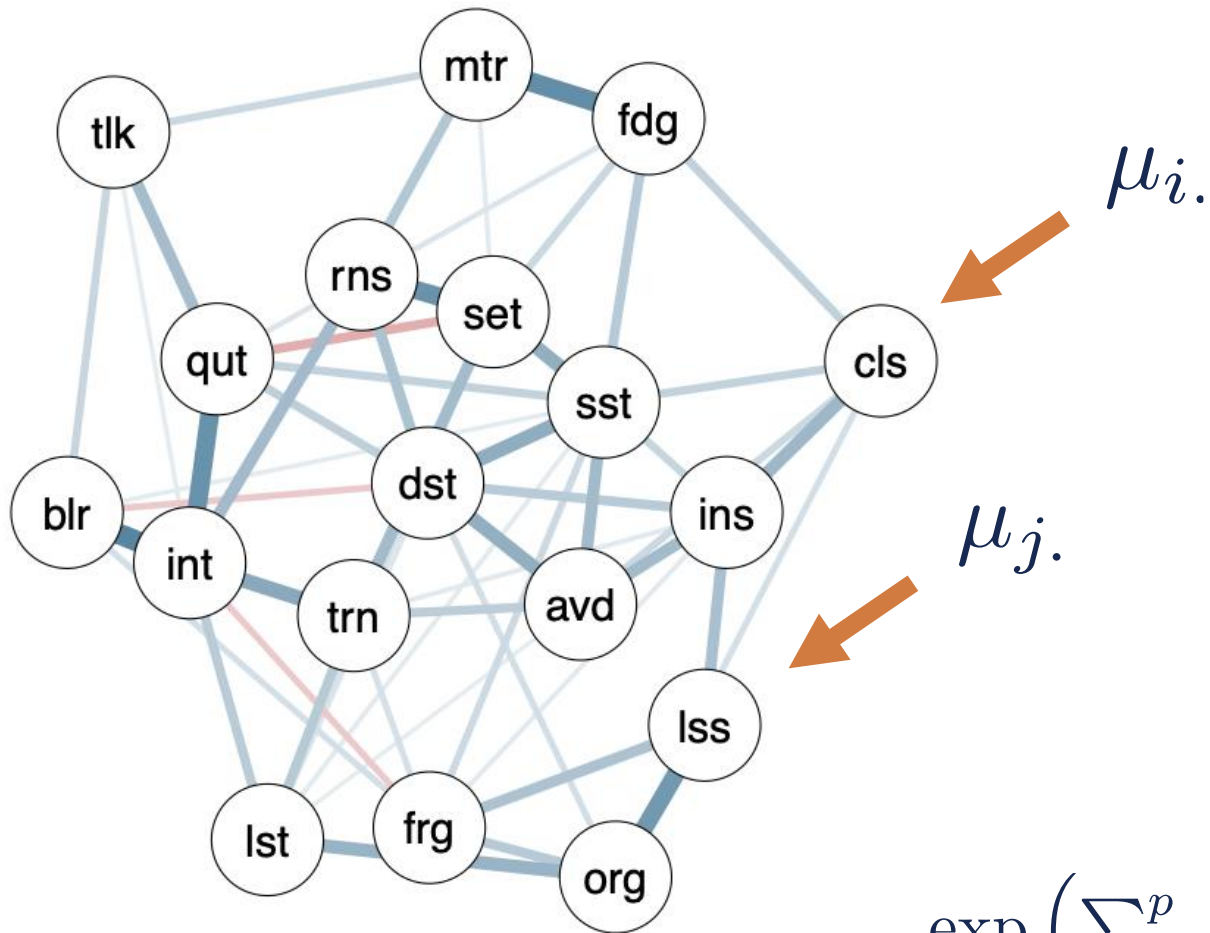
Ising Model or Ordinal Markov Random Field



Nodes are binary or ordinal variables.

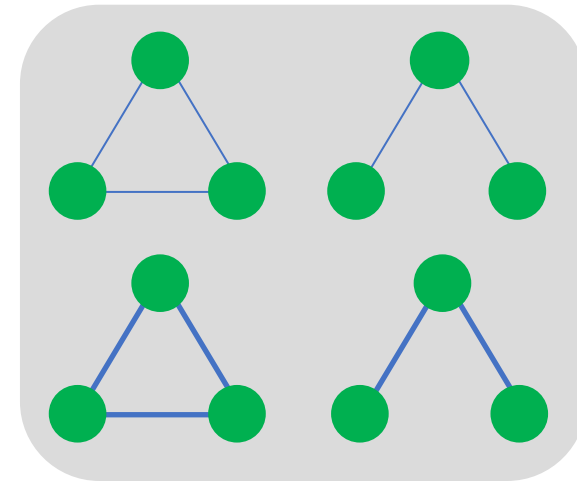
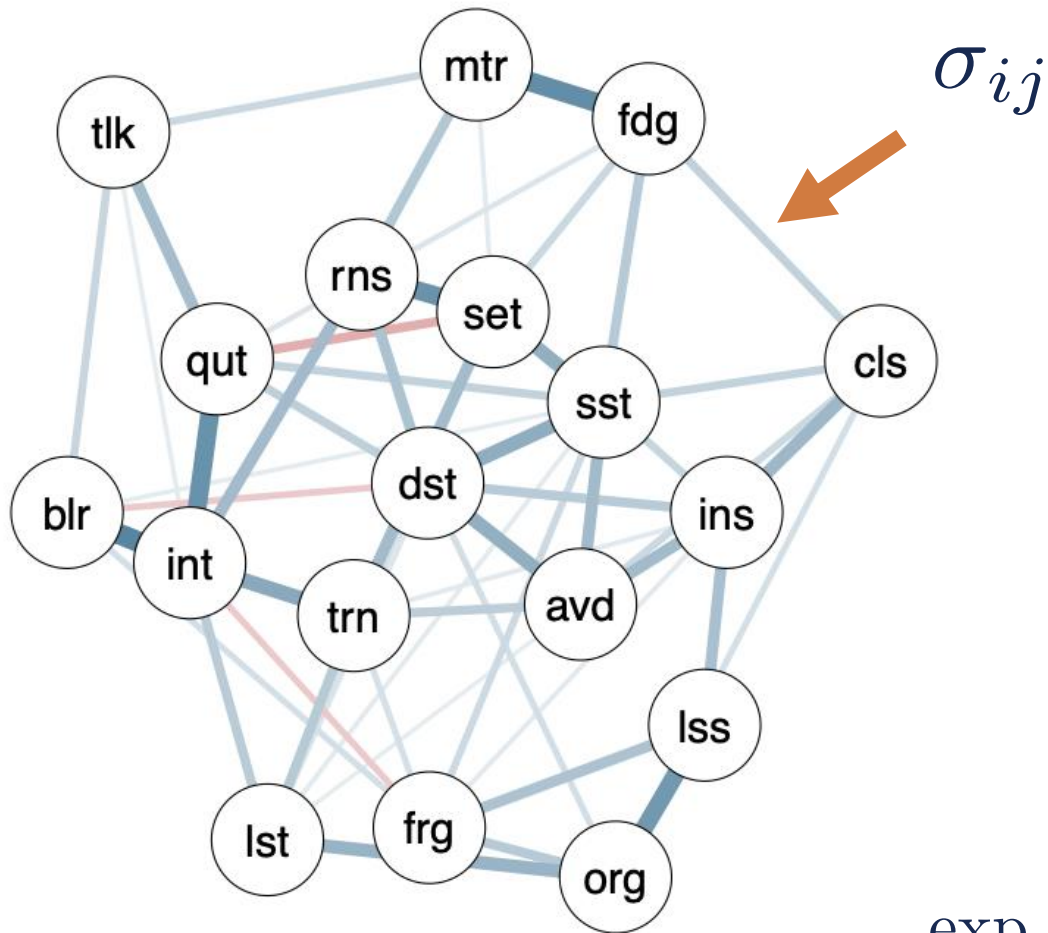
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Ising Model or Ordinal Markov Random Field



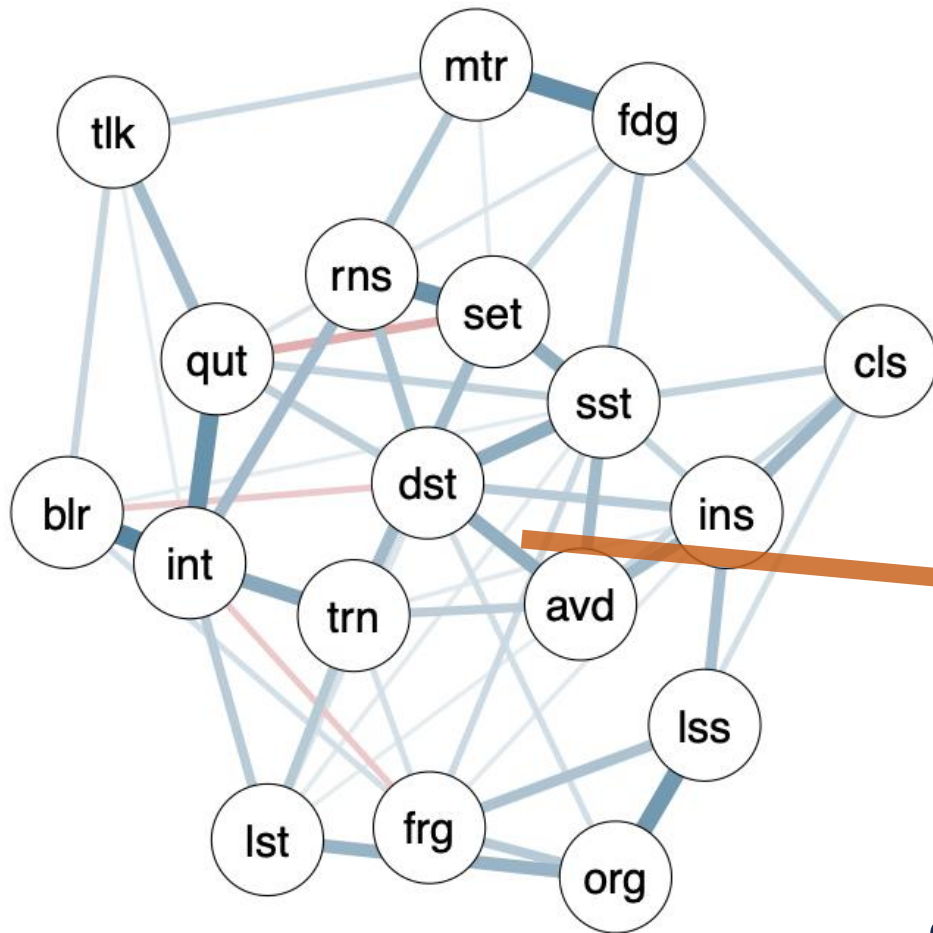
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Ising Model or Ordinal Markov Random Field

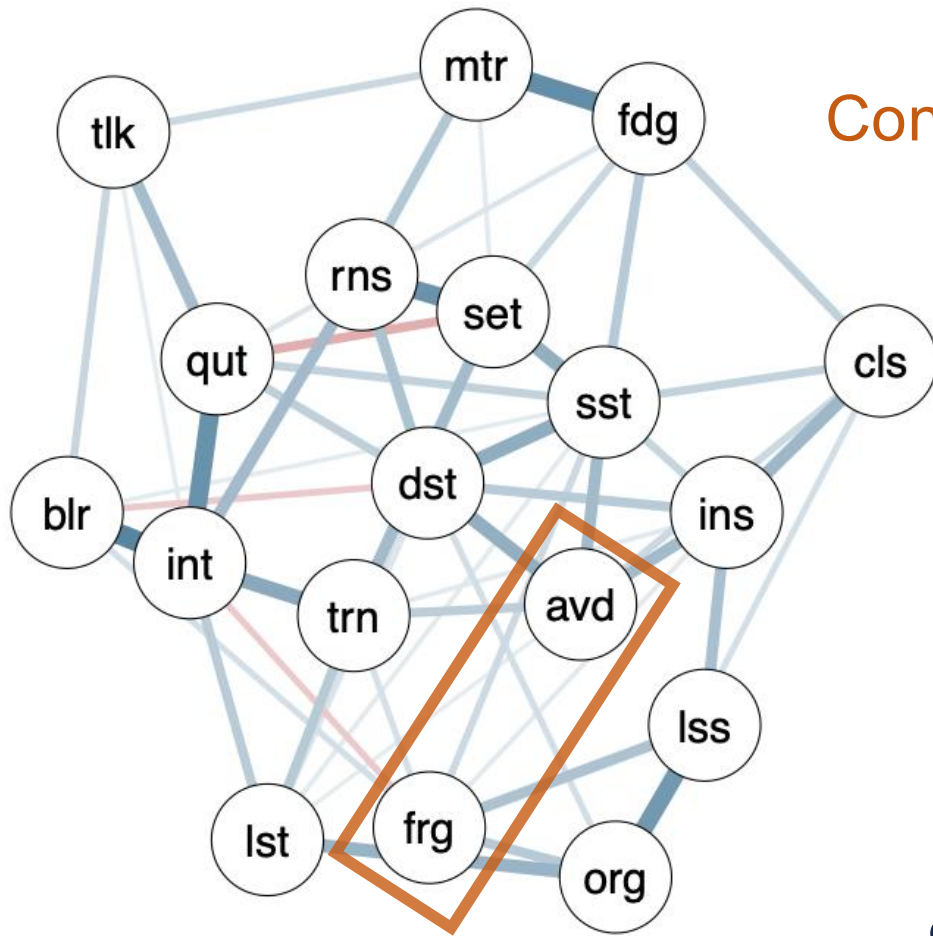


BAYESIAN ANALYSIS OF NETWORKS							
Model type: ordinal							
Number of nodes: 18							
Fitting Package: bgms							

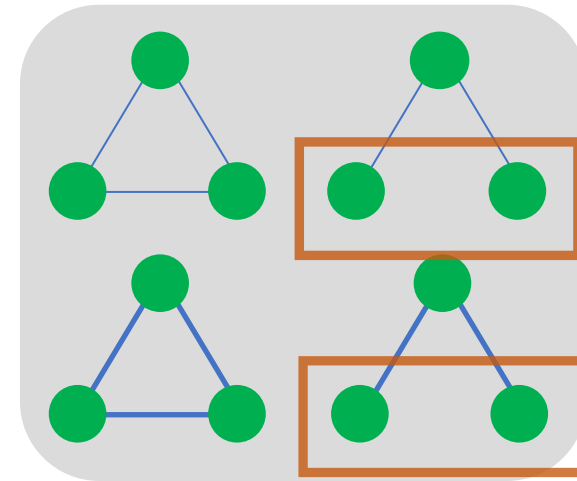
EDGE SPECIFIC OVERVIEW							
	Relation	Estimate	Posterior	Incl. Prob.	Inclusion BF	Category	Convergence
	closeatt-avoid	0.800		0.892	8.259	inconclusive	1.007
	distract-avoid	1.644		1.000	Inf	included	1.000
	forget-avoid	0.227		0.389	0.637	inconclusive	1.003

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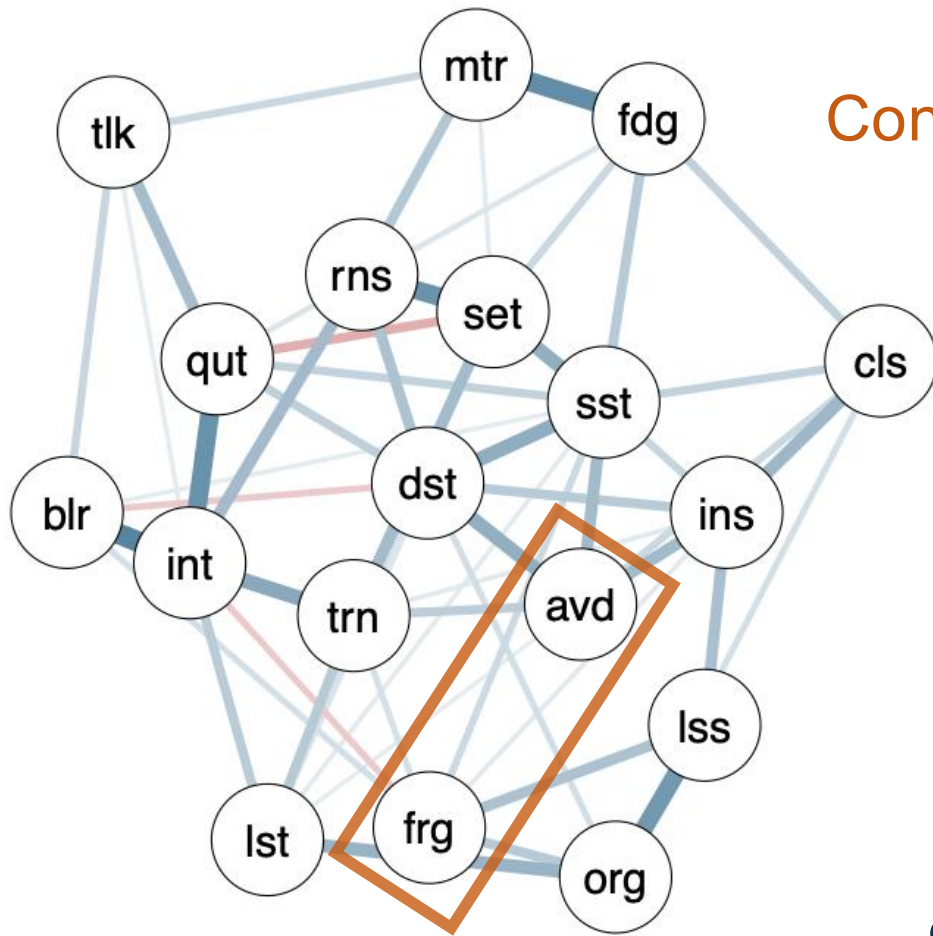


Conditional independence?



$$P(\mathbf{X} = \mathbf{x}) = \frac{\exp \left(\sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left(\sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

Ising Model or Ordinal Markov Random Field



Conditional independence?

BAYESIAN ANALYSIS OF NETWORKS

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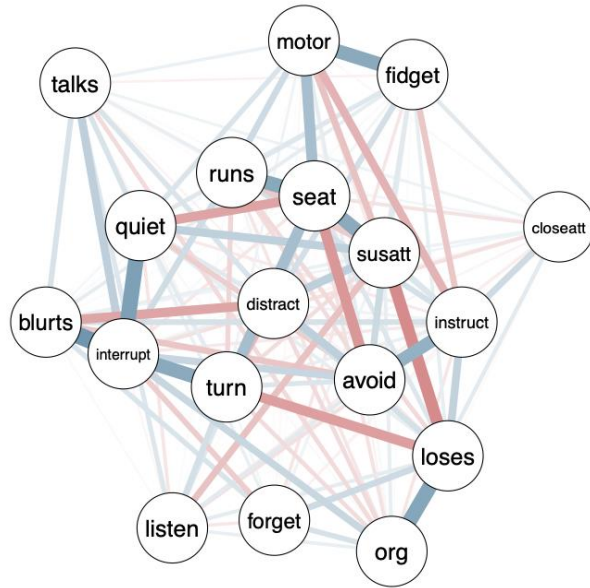
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EDGE SPECIFIC OVERVIEW

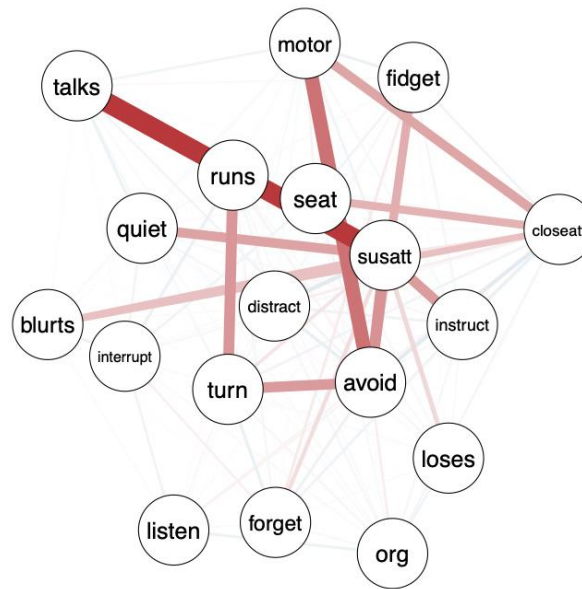
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Networks for Diagnosed and Undiagnosed



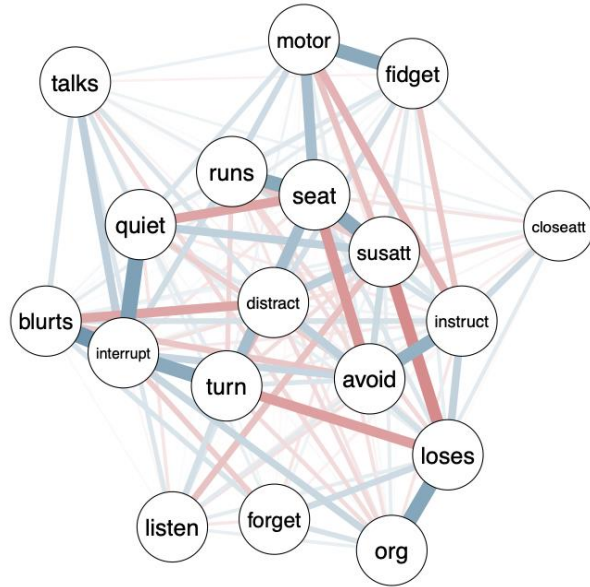
Diagnosed (n = 146)



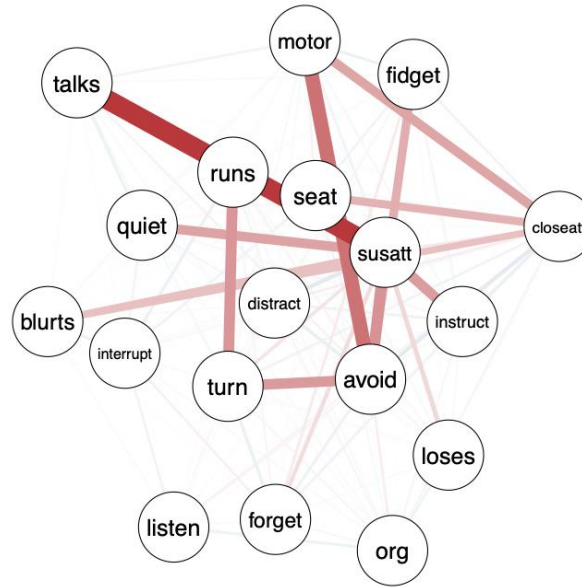
Undiagnosed (n = 209)

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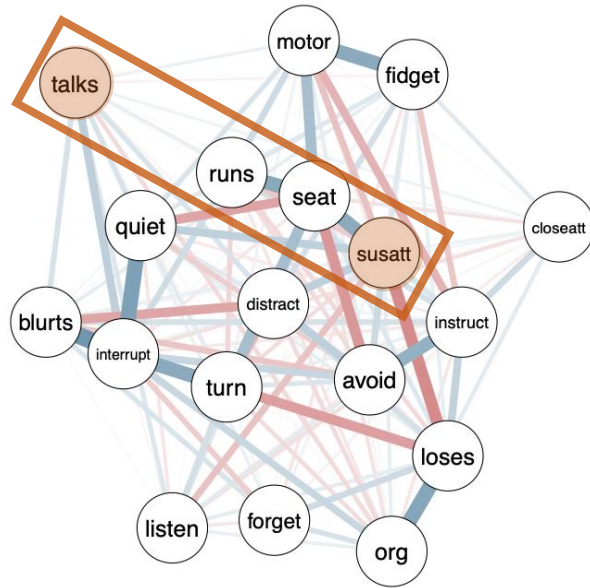


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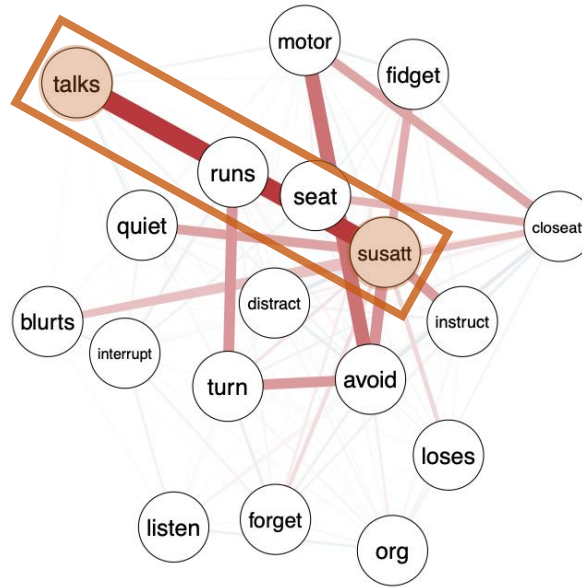
How could we assess sample differences?

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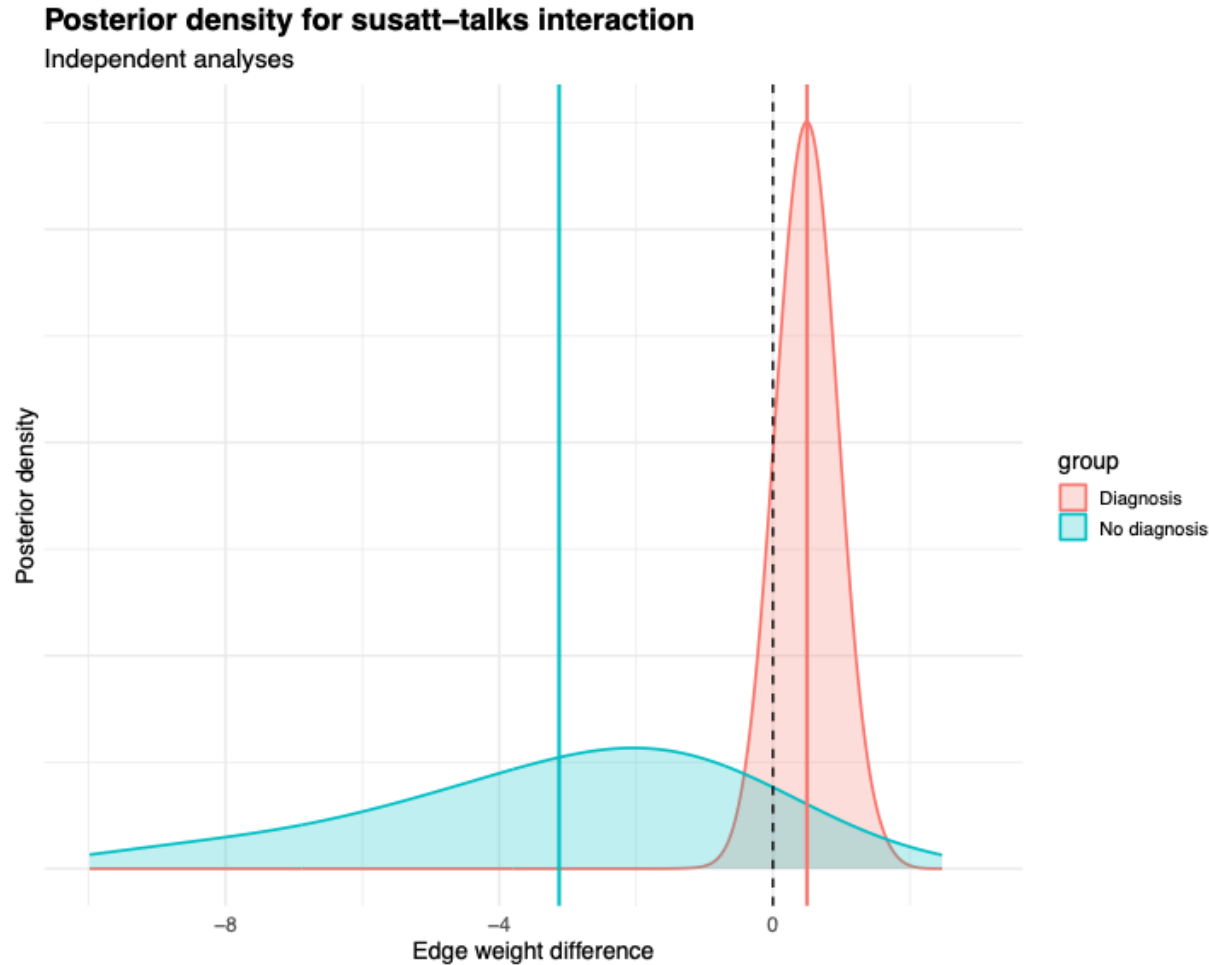


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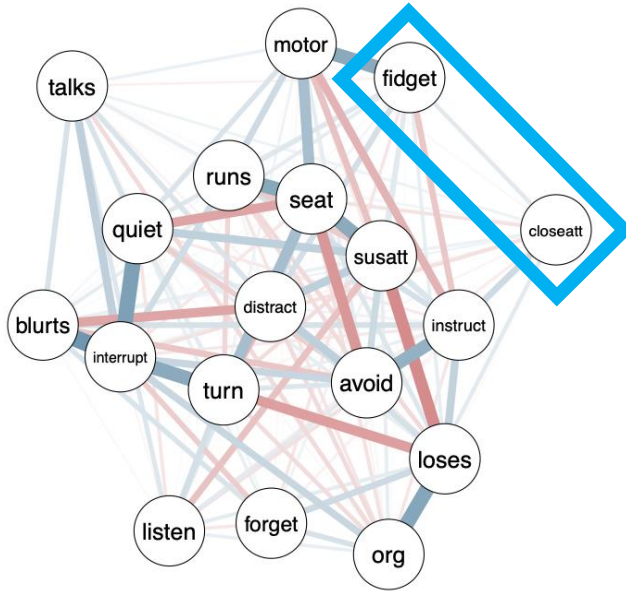


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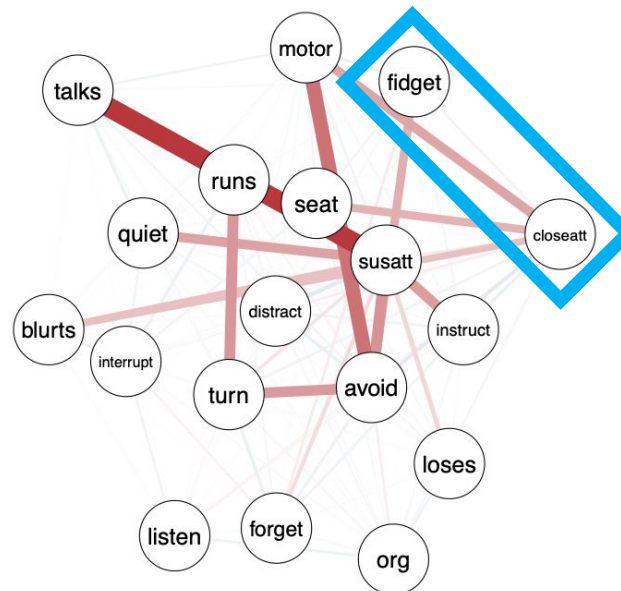
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Model the Difference in Edge Weight

$$\sigma_{ij} = \begin{cases} \phi_{ij} - \frac{1}{2}\delta_{ij} & \text{for group 1} \\ \phi_{ij} + \frac{1}{2}\delta_{ij} & \text{for group 2} \end{cases}$$



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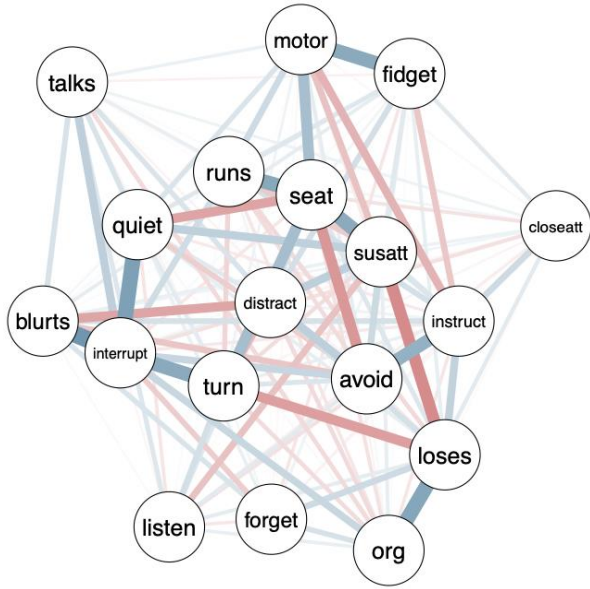


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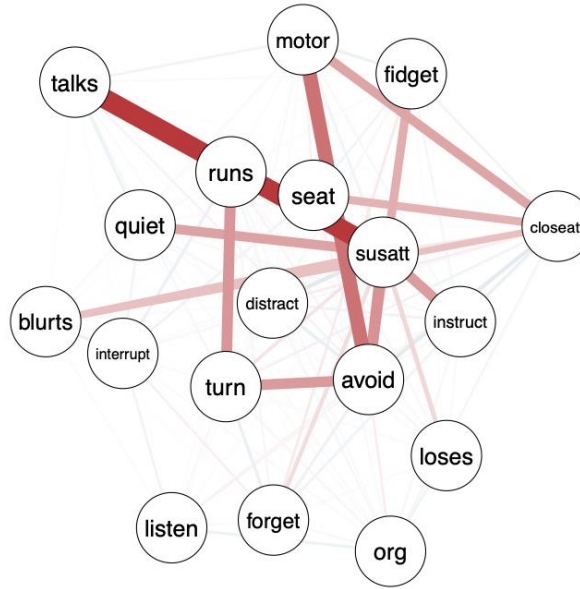
Silk, et al. (2019). A network analysis approach to ADHD symptoms: More than the sum of its parts. *PLOS ONE*, 14(1), e0211053.

Marsman, Waldorp, Sekulovski, & Haslbeck (2025). Bayes factor tests for group differences in ordinal and binary graphical models. *Psychometrika*, 90.

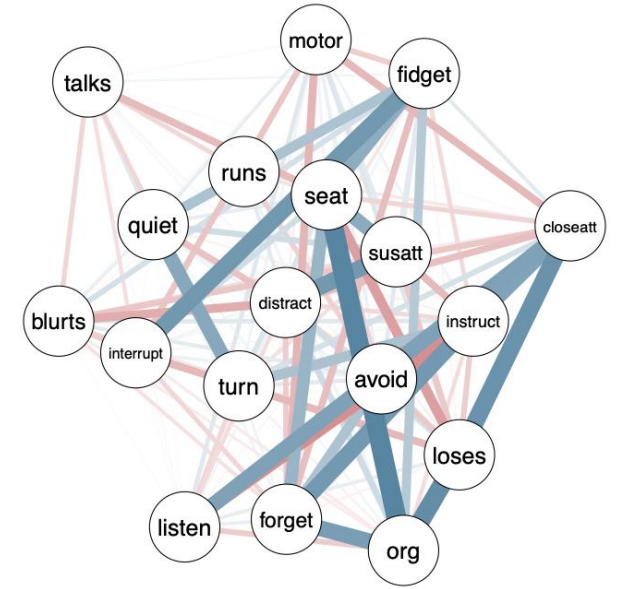
Model the Difference in Edge Weight



Diagnosed (n = 146)



Undiagnosed (n = 209)



$$\delta_{\text{diff}} = \sigma_{\text{diag}} - \sigma_{\text{no diag}}$$

Test the Difference in Edge Weight

Rather than testing whether an edge weight should be included or excluded (i.e., conditional independence), we now test if an edge difference should be included or excluded (i.e., edge invariance).

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$\mathcal{H}_0$: Edge is equivalent in the two groups $\iff \delta_{ij} = 0$

$\mathcal{H}_1$: Edge is different in the two groups $\iff \delta_{ij} \neq 0$

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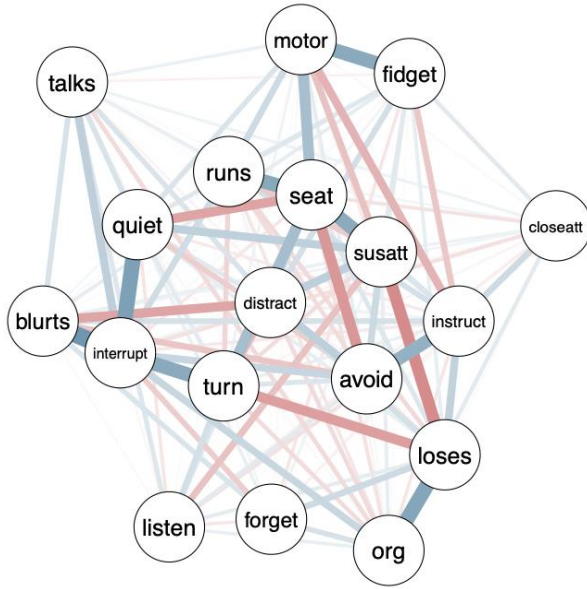
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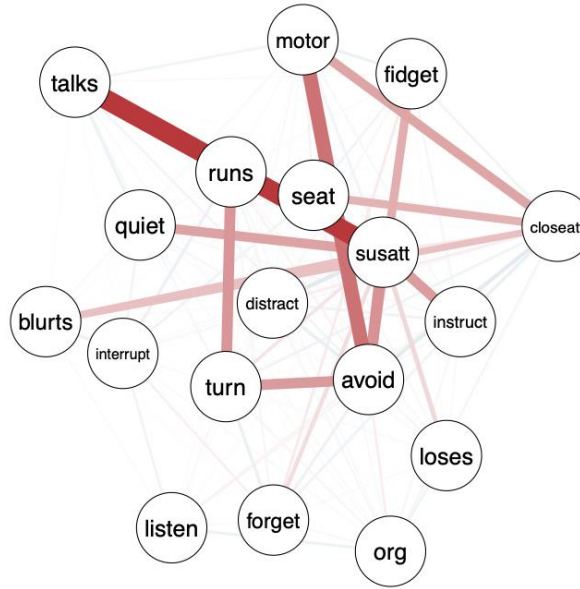
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$$\text{BF}_{10} = \frac{p(\mathcal{H}_1 \mid \text{data})}{p(\mathcal{H}_0 \mid \text{data})} \bigg/ \frac{p(\mathcal{H}_1)}{p(\mathcal{H}_0)}$$

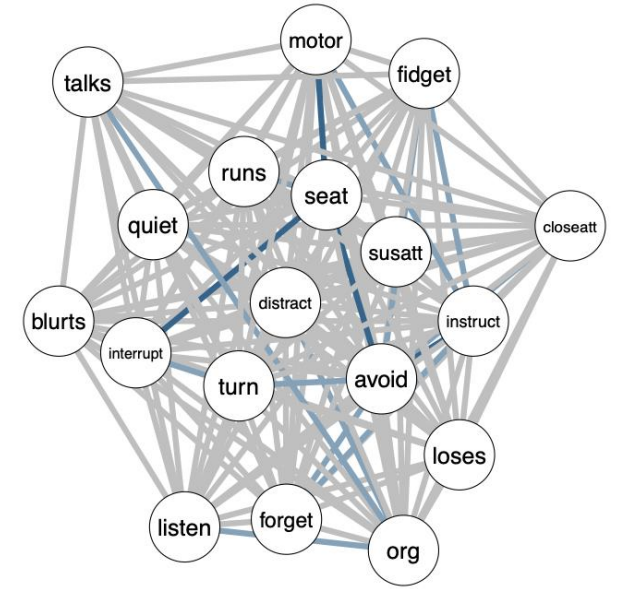
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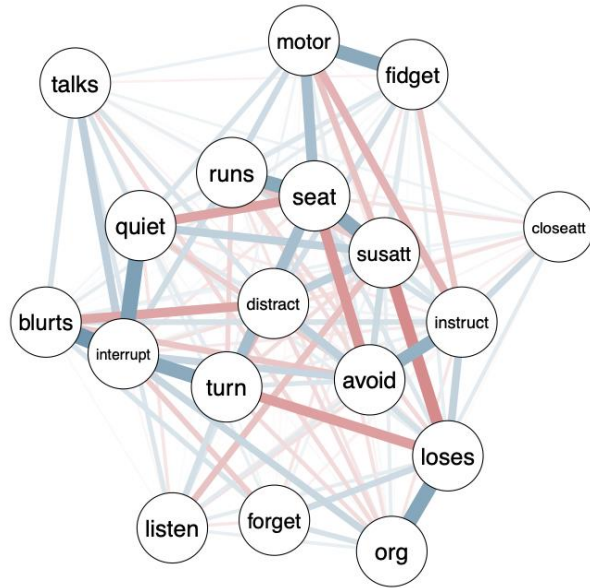
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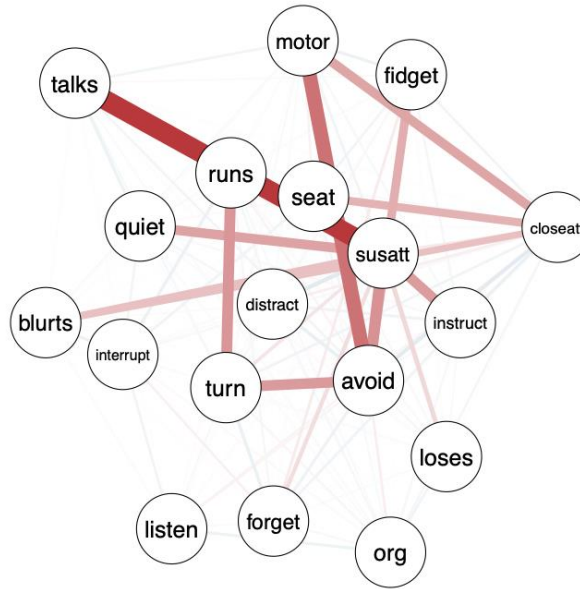
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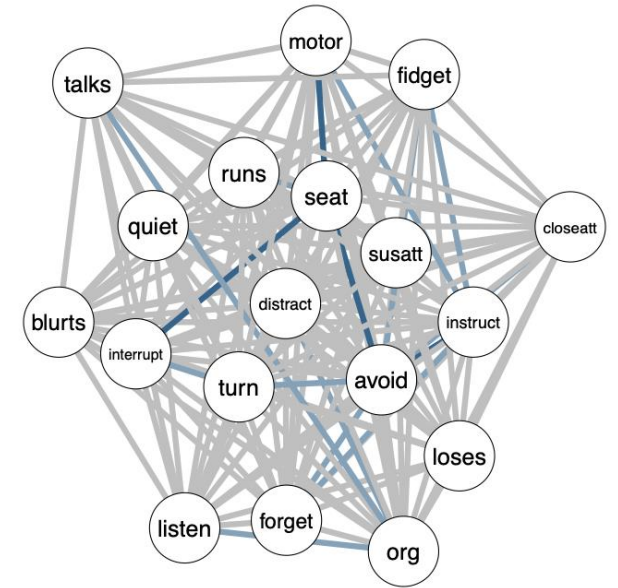
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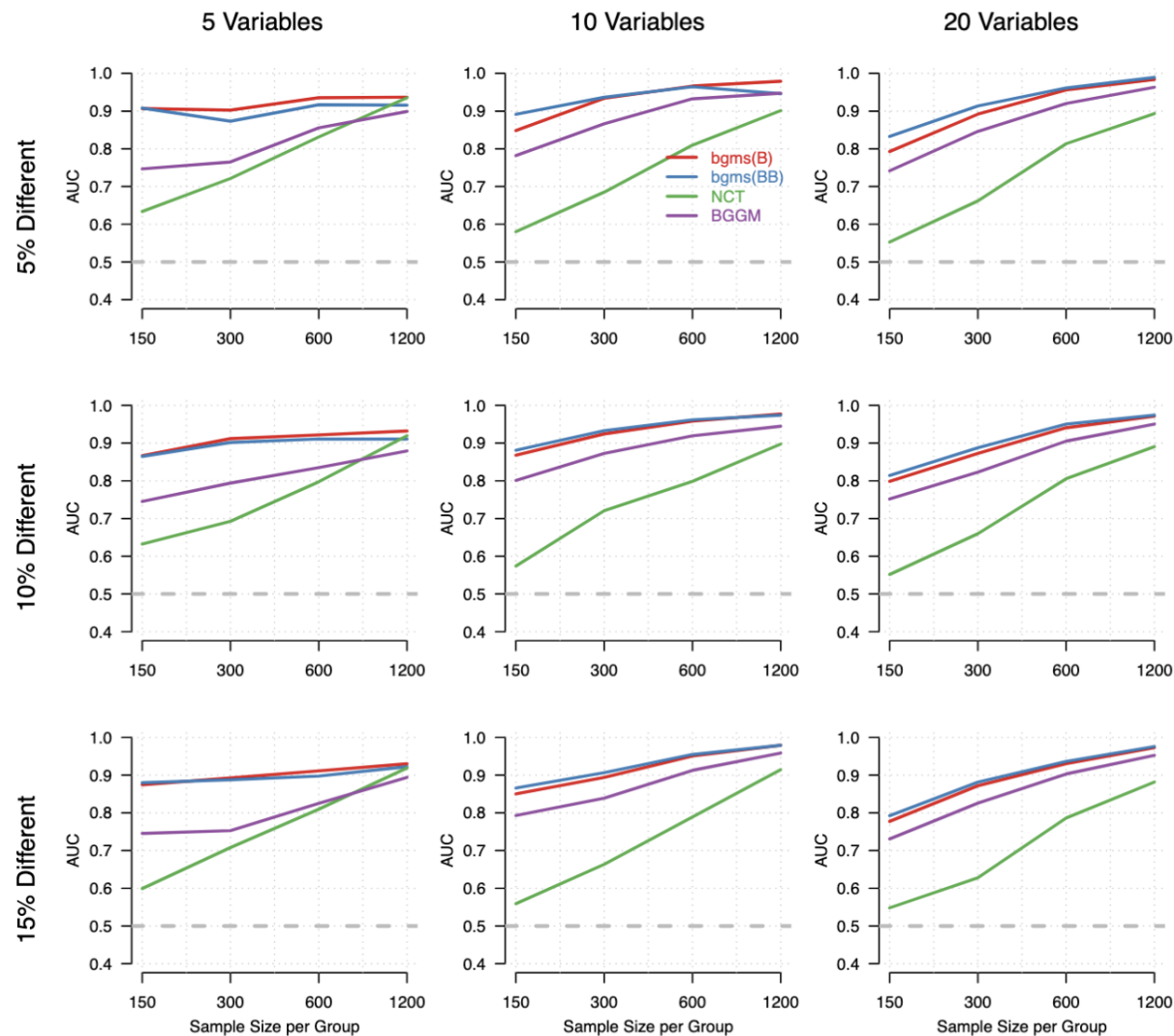


Undiagnosed (n = 209)



```
easy_fit = easybgm_compare(data = ADHD[, -1],  
                           group_indicator = ADHD[, 1],  
                           iter = 2000,  
                           package = "bgms",  
                           type = "ordinal")  
plot_edgееvidence(easy_fit, legend = FALSE,  
                  labels = colnames(ADHD[, -1]))
```

Simulation Evidence



Area under the ROC curve

- Threshold-free summary of detection accuracy
- 0.5 = chance; higher = better

Design:

- Variables: 5, 10, 20
- Edge differences: 5–15%
- Sample size: 150–1200 per group

Results:

- Bayesian MRF methods perform best overall
- Strong gains in small samples and low signal
- NCT competitive with large samples & few variables